TASK 4: CAN NETWORK AND TRAILER ELECTRICAL FAULT DIAGNOSIS

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 5 | v1.1

TIME
90 minutes (adapt to workplace conditions)

EVIDENCE
Practical | Knowledge | Documentation

PROGRAMME POINT
Weeks 59-74 | Months 16-20

Your task

Diagnose a network or trailer-electrical concern using circuit information, scan-tool data and targeted measurements. Avoid unnecessary parts replacement and document the evidence trail.

What you must do

1. Confirm the symptom, affected systems and any recent work.
2. Obtain the network topology, connector information and test limits.
3. Complete visual, supply/ground and circuit-protection checks.
4. Use scan data and targeted voltage/resistance/waveform tests as authorised.
5. Identify root cause, repair/referral action and post-repair network checks.

What good evidence looks like

SAFE CONTROL
Follow network shutdown and battery-isolation requirements; do not pierce sealed wiring or short communication circuits.

CORRECT METHOD
A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE
Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS
A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- Heavy vehicle/trailer or network rig
- Multimeter, breakout leads and optional oscilloscope
- OEM-capable scan tool
- Wiring/topology data and PPE

SOW ALIGNMENT

Block 5 - 12/24V Electrical, Electronic and Networked Systems

KSB evidence focus: K3, S6, S9; reinforced K1, K4, K10, S1-S4, S8, S10-S11, B2-B3

Task reference: HV_L3_Block_5_Task_4_CAN_Network_and_Trailer_Electrical_Fault_Diagnosis

TASK 4: CAN NETWORK AND TRAILER ELECTRICAL FAULT DIAGNOSIS

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

DTC / module status		Battery / supply voltage	
Ground voltage drop		Network resistance	
CAN high / low value		Trailer connector result	
Root cause evidence		Post-repair scan result	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why should power and grounds be checked before replacing a control unit?

2. What conditions affect a network-resistance measurement?

Knowledge check - continued

3. Explain how one wiring fault can create multiple DTCs.

4. How do you protect connectors and control units during testing?

5. What evidence confirms the whole network has recovered?

Assessor/Workplace Mentor observation

- ☐ Uses topology and diagrams to plan efficient tests.
- ☐ Checks power, ground and protection before network diagnosis.
- ☐ Selects safe back-probing/breakout methods.
- ☐ Interprets scan and measurement evidence logically.
- ☐ Verifies communication, clears/records codes appropriately.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- ☐ PASS
- ☐ REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

FAULT SCENARIOS AND EVIDENCE

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 5 | Task 4

How to use this sheet

Work through the listed scenarios during the block using a vehicle, safe fault rig or supported case study. Record which scenario was used for this task and reference the actual measurements and assessor observation on the existing practical record. A discussion or case study alone is not evidence of practical competence.

1. Brake lights inoperative

Checks: Confirm all brake lamps, switch/input, protection, supply, ground and body/trailer control outputs.

Evidence: Record circuit readings, corrected cause and lamp function.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

2. Rear fog light inoperative

Checks: Check switch request, feed, fuse, connector, lamp and permitted trailer interactions against the diagram.

Evidence: Record test points and final rear fog function.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

3. Nearside headlamp dim

Checks: Compare both sides under load; measure feed and ground voltage drop and inspect connector, lamp and earth.

Evidence: Record side-by-side voltage values and restored output.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference:

4. Offside sidelights inoperative

Checks: Identify affected lamps and shared branch; check fuse, control output, connector, continuity and ground.

Evidence: Record failed point, repair and operational check.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

Scenario selected for the original task assessment: _____

For any safety-critical fault, stop or control the work and follow the manufacturer, DVSA and workplace procedures. Use the original task pages for findings, feedback, PASS/REFER and signatures.

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.